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Project 4 report

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This project develops a workflow for converting spectral reflectance data into CIE XYZ and CIELAB values and then measuring perceptual color differences with Delta E*ab.

- Step 2 uses a vectorized `ref2XYZ` function so multiple reflectance spectra can be converted to XYZ simultaneously.
- Step 3 tests that conversion with the 24 patches of a ColorChecker under CIE illuminant D65 and the CIE 1931 2-degree standard observer.
- Steps 4 and 5 introduce `XYZ2Lab` and use the D65 perfect-diffuser white point to convert the ColorChecker XYZ values to CIELAB.
- Step 6 repeats the conversion after reducing the ColorChecker reflectances to 2 percent of their original values. This demonstrates CIELAB behavior for very dark samples.
- Steps 7 and 8 introduce the CIE 1976 Delta E*ab calculation and compare the ColorChecker with a MetaChecker under D65 and illuminant A. The MetaChecker contains different reflectance spectra designed to produce similar colors under one illuminant but different colors under another, demonstrating illuminant metamerism.
- Step 9 applies the same XYZ, CIELAB, and Delta E*ab calculations to the physical/real, imaged, and visually matched paint-patch measurements collected in Project 2.
- Step 10 plots the paint-patch results in the CIELAB a*-b* plane, including reference circles that provide visual context for the color differences.

```
repo_dir = fileparts(fileparts(fileparts(fileparts(mfilename('fullpath')))));
addpath(fullfile(repo_dir, 'shared'));
paths = imgs351Paths(repo_dir);
addpath(paths.p03.code, paths.p04.code);
```

step 2 - modify function ref2XYZ

Modify the `ref2XYZ` function so it can simultaneously process multiple reflectance spectra (`ref(s)`).

```
function XYZ = ref2XYZ(reflectances, colorMatchingFunctions, illuminant)
%REF2XYZ Convert reflectance spectra to CIE XYZ tristimulus values.
% XYZ = REF2XYZ(REFLECTANCES, COLORMATCHINGFUNCTIONS, ILLUMINANT)
% accepts one or more reflectance spectra in an N-by-M matrix, the CIE
% color-matching functions in an N-by-3 matrix, and an N-by-1 illuminant
% spectral power distribution. The result is a 3-by-M matrix scaled so a
% perfect reflecting diffuser has Y = 100.

normalization = 100 / sum(colorMatchingFunctions(:, 2) .* illuminant);
weightedReflectances = illuminant .* reflectances;
XYZ = normalization * colorMatchingFunctions.' * weightedReflectances;
end
```

step 3 - test function ref2XYZ

test the modified `ref2XYZ` function with ColorChecker reflectance values

```
% load CIE data
cie = loadCIEdata;

% load ColorChecker spectra data
CC_spectra = load(fullfile(paths.p04.data, ...
    'ColorChecker_380-780-5nm.txt'));

% calculate XYZ values for ColorChecker patches
CC_XYZs = ref2XYZ(CC_spectra(:, 2:25), cie.cmf2deg, cie.illD65);

% display ColorChecker XYZ values
CC_XYZs
```

CC_XYZs =						
Columns 1 through 7						
11.5145	39.1346	18.3488	11.1492	25.8437	31.7110	37.1457
10.3819	36.5981	19.6332	13.8551	24.3868	43.8600	29.5592

ColorChecker XYZ and Lab values (D65 illuminant and 2 deg. observer)								
Patch	#	X	Y	Z	L*	a*	b*	Patch Name
1	11.515	10.382	7.150	38.519	12.410	13.309		Dark Skin
2	39.135	36.598	27.956	66.974	14.329	17.320		Light Skin
3	18.349	19.633	35.647	51.420	-1.624	-21.603		Blue Sky
4	11.149	13.855	7.427	44.024	13.963	21.774		Foliage
5	25.844	24.387	45.614	56.473	11.544	-24.698		Blue Flower
6	31.711	43.860	44.878	72.135	-33.101	3.115		Bluish Green
7	37.146	29.559	6.501	61.272	32.497	55.059		Orange
8	13.863	12.318	39.309	41.717	14.416	-42.900		Purplish Blue
9	29.133	19.847	14.994	51.664	45.468	13.382		Moderate Red
10	8.589	6.457	15.474	30.537	23.785	-24.136		Purple
11	33.917	44.153	11.430	72.331	-26.083	57.948		Yellow Green
12	46.186	42.496	8.677	71.211	17.187	64.297		Orange Yellow
13	8.918	6.418	32.274	30.443	27.024	-53.277		Blue

14	15.035	24.108	9.638	56.196	-40.771	35.342	Green
15	19.345	11.358	5.553	40.176	51.976	22.689	Red
16	55.846	58.973	9.641	81.277	-0.508	78.575	Yellow
17	29.677	19.352	32.263	51.096	50.004	-17.653	Magenta
18	14.414	19.975	39.001	51.809	-25.642	-25.126	Cyan
19	87.840	92.378	95.613	96.975	0.076	3.262	White
20	57.962	61.043	65.491	82.402	-0.133	0.831	Neutral 8
21	35.229	37.041	40.226	67.308	0.079	0.125	Neutral 6.5
22	19.349	20.471	22.154	52.365	-0.541	0.237	Neutral 5
23	8.765	9.291	10.319	36.540	-0.568	-0.600	Neutral 3.5
24	3.211	3.376	3.931	21.492	0.035	-1.462	Black

step 6 - test XYZ2Lab function using darker ColorChecker

```
% multiply all the ColorChecker spectra by 0.02
CC_spectra_darker = CC_spectra .* 0.02;

% calculate the XYZ values for darker ColorChecker
CC_XYZs_darker = ref2XYZ(CC_spectra_darker(:, 2:25), cie.cmf2deg, cie.illD65);

% calculate the Lab values for darker ColorChecker
CC_Labs_darker = XYZ2Lab(CC_XYZs_darker, XYZn_D65);

% read in the names of the ColorChecker patches
names = cellstr(readlines(fullfile(paths.p04.data, ...
    'ColorChecker_names.txt')));

% print the formatted table
% header
fprintf('\n%s\n', "ColorChecker(Dark) XYZ and Lab values (D65 illuminant and 2 deg. observer)");
fprintf('%5s %2s %5s %7s %7s %7s %7s %-13s\n', 'Patch', '#', ...
    'X', 'Y', 'Z', 'L*', 'a*', 'b*', 'Patch Name');
% loop to print the patch values
for i = 1:size(CC_Labs, 2)
    fprintf('      %2d %7.3f %7.3f %7.3f %7.3f %7.3f %7.3f %-13s\n', i, ...
        CC_XYZs_darker(1,i), CC_XYZs_darker(2,i), CC_XYZs_darker(3,i), ...
        CC_Labs_darker(1,i), CC_Labs_darker(2,i), CC_Labs_darker(3,i), names{i});
end
```

Patch	#	X	Y	Z	L*	a*	b*	Patch Name
1	0.230	0.208	0.143	1.876	1.350	1.188	Dark Skin	
2	0.783	0.732	0.541	6.612	3.565	3.659	Light Skin	
3	0.367	0.393	0.713	3.547	-0.255	-4.082	Blue Sky	
4	0.223	0.277	0.149	2.503	-1.654	2.191	Foliage	
5	0.517	0.488	0.912	4.406	2.184	-5.453	Blue Flower	
6	0.634	0.877	0.898	7.924	-8.173	0.823	Bluish Green	
7	0.743	0.591	0.130	5.340	7.416	7.347	Orange	
8	0.277	0.246	0.786	2.225	1.766	-7.409	Purplish Blue	
9	0.583	0.397	0.300	3.586	8.414	1.893	Moderate Red	
10	0.172	0.129	0.309	1.166	2.009	-2.416	Purple	
11	0.678	0.883	0.229	7.977	-6.593	10.483	Yellow Green	
12	0.924	0.850	0.174	7.677	4.646	10.754	Orange Yellow	
13	0.178	0.128	0.645	1.159	2.309	-7.234	Blue	
14	0.301	0.482	0.193	4.355	-6.454	4.752	Green	
15	0.387	0.227	0.111	2.052	7.005	1.949	Red	
16	1.117	1.179	0.193	10.405	-0.138	15.181	Yellow	
17	0.594	0.387	0.645	3.496	9.246	-3.202	Magenta	
18	0.288	0.399	0.780	3.609	-3.745	-4.935	Cyan	
19	1.757	1.848	1.912	14.666	0.021	0.885	White	
20	1.159	1.221	1.310	10.710	-0.036	0.226	Neutral 8	
21	0.705	0.741	0.805	6.692	0.019	0.030	Neutral 6.5	
22	0.387	0.409	0.443	3.698	-0.088	0.038	Neutral 5	
23	0.175	0.186	0.206	1.679	-0.054	-0.058	Neutral 3.5	
24	0.064	0.068	0.079	0.610	0.002	-0.073	Black	

step 7 - create function deltaEab

Create a function that takes 2 sets of CIELAB values and return the DEab values

```
function deltaE = deltaEab(Lab1, Lab2)
%DELTAEAB Calculate CIE 1976 Delta E*ab color differences.
% DELTAE = DELTAEAB(LAB1, LAB2) accepts corresponding CIELAB samples in
% two 3-by-N matrices and returns their Euclidean distances as a 1-by-N
% row vector.

deltaE = sqrt(sum((Lab2 - Lab1) .^ 2, 1));
end
```

step 8 - test deltaEab function

```

% calculate the XYZ of the standard ColorChecker and the MetaChecker
% under illuminant D65 and illuminant A

% compute the XYZ values of A
XYZn_A = ref2XYZ(cie.PRD, cie.cmf2deg, cie.illA);

% ColorChecker XYZ value under D65
CC_XYZs_D65 = CC_XYZs;
% calculate ColorChecker XYZ value under A
CC_XYZs_A = ref2XYZ(CC_spectra(:, 2:25), cie.cmf2deg, cie.illA);
% CIE Lab values of ColorChecker under D65
CC_Labs_D65 = CC_Labs;
% calculate the CIE Lab values of ColorChecker under A
CC_Labs_A = XYZ2Lab(CC_XYZs_A, XYZn_A);

% load MetaChecker spectra data
MetaCC_spectra = load(fullfile(paths.p04.data, ...
    'MetaChecker_380-780-5nm.txt'));
% calculate MetaChecker XYZ value under D65
MetaCC_XYZs_D65 = ref2XYZ(MetaCC_spectra(:, 2:25), cie.cmf2deg, cie.illD65);
% calculate MetaChecker XYZ value under A
MetaCC_XYZs_A = ref2XYZ(MetaCC_spectra(:, 2:25), cie.cmf2deg, cie.illA);
% calculate the Lab values of MetaChecker under D65
MetaCC_Labs_D65 = XYZ2Lab(MetaCC_XYZs_D65, XYZn_D65);
% calculate the Lab values of MetaChecker under A
MetaCC_Labs_A = XYZ2Lab(MetaCC_XYZs_A, XYZn_A);

% compute the color difference under D65
deltaEab_D65 = deltaEab(CC_Labs_D65, MetaCC_Labs_D65);
% compute the color difference under A
deltaEab_A = deltaEab(CC_Labs_A, MetaCC_Labs_A);

% print the color difference under different illuminants
% header
fprintf('\n%s\n', 'ColorChecker and MetaChecker color difference');
fprintf('%5s %2s %9s %9s\n', 'Patch', '#', 'DEab(D65)', 'DEab(illA)');
% loop to print the patch values
for i = 1:size(deltaEab_D65, 2)
    fprintf('      %2d %9.3e %6.3f\n', i, deltaEab_D65(i), deltaEab_A(i));
end

```

ColorChecker and MetaChecker color difference

Patch	#	DEab(D65)	DEab(illA)
	1	2.597e-07	22.636
	2	1.136e-07	22.178
	3	1.056e-07	32.275
	4	1.905e-07	28.232
	5	3.980e-07	25.937
	6	1.326e-07	29.487
	7	8.581e-08	17.309
	8	1.454e-07	27.241
	9	1.665e-07	12.210
	10	2.907e-07	19.509
	11	1.561e-07	22.623
	12	1.305e-07	16.970
	13	1.083e-07	20.083
	14	1.193e-07	26.099
	15	6.708e-08	7.053
	16	1.330e-07	11.532
	17	6.468e-09	10.690
	18	8.581e-08	31.619
	19	2.661e-07	2.545
	20	6.948e-08	15.940
	21	1.846e-07	28.926
	22	8.337e-08	26.751
	23	3.668e-07	20.574
	24	1.022e-07	18.567

step 9 - calculate CIE Lab and color difference for paint patches

```

% define ColorMunki/Argyll/spotread measurement wavelengths
cm_lams = 380:10:730;

% define header offsets for reading the .sp files
cm_g_offset_spotread = 19;
cm_g_offset_spotreadt = 18;

% load and normalize the measured spectral data for the patch 3.1
data_real31 = importdata(fullfile(paths.p02.data, 'colormunki_data', ...
    'real', '3.1_real.sp'), ' ', ...
    cm_g_offset_spotread);
real_31 = data_real31.data/100;

```

```

data_imaged31 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'imaged', ...
        '3.1_imaged.sp'), ' ', ...
    cm_g_offset_spotreadt);
imaged_31 = data_imaged31.data/100;

data_matching31 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'matched', ...
        '3.1_matching.sp'), ' ', ...
    cm_g_offset_spotreadt);
matching_31 = data_matching31.data/100;

% load and normalize the measured spectral data for the patch 3.2
data_real32 = importdata(fullfile(paths.p02.data, 'colormunki_data', ...
    'real', '3.2_real.sp'), ' ', ...
    cm_g_offset_spotreadt);
real_32 = data_real32.data/100;

data_imaged32 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'imaged', ...
        '3.2_imaged.sp'), ' ', ...
    cm_g_offset_spotreadt);
imaged_32 = data_imaged32.data/100;

data_matching32 = importdata( ...
    fullfile(paths.p02.data, 'colormunki_data', 'matched', ...
        '3.2_matching.sp'), ' ', ...
    cm_g_offset_spotreadt);
matching_32 = data_matching32.data/100;

% INTERPOLATE/EXTRAPOLATE DATA

interpolated_real_31 = interp1(cm_lams, transpose(real_31), cie.lambda(:), 'linear', ...
    'extrap');
interpolated_imaged_31 = interp1(cm_lams, transpose(imaged_31), cie.lambda(:), ...
    'linear', 'extrap');
interpolated_matching_31 = interp1(cm_lams, transpose(matching_31), cie.lambda(:), ...
    'linear', 'extrap');

interpolated_real_32 = interp1(cm_lams, transpose(real_32), cie.lambda(:), 'linear', ...
    'extrap');
interpolated_imaged_32 = interp1(cm_lams, transpose(imaged_32), cie.lambda(:), ...
    'linear', 'extrap');
interpolated_matching_32 = interp1(cm_lams, transpose(matching_32), cie.lambda(:), ...
    'linear', 'extrap');

% CALCULATING XYZ

% Calculate the tristimulus values and chromaticity coordinates
% for patch 3.1
XYZ_real_31 = ref2XYZ(interpolated_real_31, cie.cmf2deg, cie.illD50);
% XYZ_real_31 = reshape(XYZ_real_31, 3, 1);

XYZ_imaged_31 = ref2XYZ(interpolated_imaged_31, cie.cmf2deg, cie.illD50);
% XYZ_imaged_31 = reshape(XYZ_imaged_31, 3, 1);

XYZ_matching_31 = ref2XYZ(interpolated_matching_31, cie.cmf2deg, cie.illD50);
% XYZ_matching_31 = reshape(XYZ_matching_31, 3, 1);

% Calculate the tristimulus values and chromaticity coordinates
% for patch 3.2
XYZ_real_32 = ref2XYZ(interpolated_real_32, cie.cmf2deg, cie.illD50);
% XYZ_real_32 = reshape(XYZ_real_32, 3, 1);

XYZ_imaged_32 = ref2XYZ(interpolated_imaged_32, cie.cmf2deg, cie.illD50);
% XYZ_imaged_32 = reshape(XYZ_imaged_32, 3, 1);

XYZ_matching_32 = ref2XYZ(interpolated_matching_32, cie.cmf2deg, cie.illD50);
% XYZ_matching_32 = reshape(XYZ_matching_32, 3, 1);

% CALCULATE CIELab

% compute the XYZ values of D50
XYZn_D50 = ref2XYZ(cie.PRd, cie.cmf2deg, cie.illD50);

% patch 3.1
Lab_real_31 = XYZ2Lab(XYZ_real_31, XYZn_D50);
Lab_imaged_31 = XYZ2Lab(XYZ_imaged_31, XYZn_D50);
Lab_matching_31 = XYZ2Lab(XYZ_matching_31, XYZn_D50);

% patch 3.2
Lab_real_32 = XYZ2Lab(XYZ_real_32, XYZn_D50);

```

```

Lab_imaged_32 = XYZ2Lab(XYZ_imaged_32, XYZn_D50);
Lab_matching_32 = XYZ2Lab(XYZ_matching_32, XYZn_D50);

% CALCULATE DELTA E AB

% compute the color difference for patch 3.1
deltaEab_imaged_31 = deltaEab(Lab_real_31, Lab_imaged_31);
deltaEab_matching_31 = deltaEab(Lab_real_31, Lab_matching_31);

% compute the color difference for patch 3.2
deltaEab_imaged_32 = deltaEab(Lab_real_32, Lab_imaged_32);
deltaEab_matching_32 = deltaEab(Lab_real_32, Lab_matching_32);

% LISTING MEASURED AND CALCULATED XYZ

% print title
fprintf('\n%s\n', "Calculated XYZ, Lab, and deltaE values (w.r.t. real patches)")

% print first table for patch 3.1
fprintf('\n
                                %8s\n', 'patch 3.1')
fprintf('%5s %8s %8s %8s %8s %8s %8s %8s\n', '', 'X', 'Y', 'Z', 'L', 'a', 'b', 'dEab')
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', 'real', XYZ_real_31, Lab_real_31)
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', 'imaged', XYZ_imaged_31, Lab_imaged_31, deltaEab_imaged_31)
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', 'matching', XYZ_matching_31, Lab_matching_31, deltaEab_matching_31)

% print second table for patch 3.2
fprintf('\n
                                %8s\n', 'patch 3.2')
fprintf('%5s %8s %8s %8s %8s %8s %8s %8s\n', '', 'X', 'Y', 'Z', 'L', 'a', 'b', 'dEab')
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', 'real', XYZ_real_32, Lab_real_32)
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', 'imaged', XYZ_imaged_32, Lab_imaged_32, deltaEab_imaged_32)
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', 'matching', XYZ_matching_32, Lab_matching_32, deltaEab_matching_32)

```

Calculated XYZ, Lab, and deltaE values (w.r.t. real patches)

	patch 3.1						
	X	Y	Z	L	a	b	dEab
real	42.1870	41.3717	25.3597	70.4356	7.0179	14.0651	
imaged	44.1485	43.6078	28.4532	71.9656	6.2170	11.4243	3.1553
matching	85.2451	87.2978	57.3463	94.8645	2.0231	13.9997	24.9344

	patch 3.2						
	X	Y	Z	L	a	b	dEab
real	32.3267	34.2556	31.9201	65.1650	-2.4970	-5.7798	
imaged	63.8858	65.2197	49.3682	84.5968	2.2903	4.9253	22.6961
matching	44.0068	42.7391	21.1241	71.3775	8.3389	23.6651	31.9846

step 10 - visualize color difference of paint patches

```

figure;
hold on
grid on

% patch 3.1
plot(Lab_real_31(2,1), Lab_real_31(3,1), 'o', 'MarkerFaceColor', 'b', ...
     'Color', 'b', 'MarkerSize', 4)
plot(Lab_imaged_31(2), Lab_imaged_31(3), 'square', 'MarkerFaceColor', 'b', ...
     'Color', 'b', 'MarkerSize', 4)
plot(Lab_matching_31(2), Lab_matching_31(3), 'diamond', 'MarkerFaceColor', 'b', ...
     'Color', 'b', 'MarkerSize', 4)

% patch 3.2
plot(Lab_real_32(2), Lab_real_32(3), 'o', 'MarkerFaceColor', 'r', 'Color', ...
     'r', 'MarkerSize', 4)
plot(Lab_imaged_32(2), Lab_imaged_32(3), 'square', 'MarkerFaceColor', 'r', 'Color', ...
     'r', 'MarkerSize', 4)
plot(Lab_matching_32(2), Lab_matching_32(3), 'diamond', 'MarkerFaceColor', 'r', 'Color', ...
     'r', 'MarkerSize', 4)

% JND reference circles for the Lab of the real patches
axis square
viscircles([Lab_real_31(2) Lab_real_31(3)], 2.5, 'Color', 'b', 'LineWidth', 0.5);
viscircles([Lab_real_32(2), Lab_real_32(3)], 2.5, 'Color', 'r', 'LineWidth', 0.5);

fontsize(gcf, scale=0.9)

hold off

xlim([-60 60])
ylim([-60 60])
xticks(-60:10:60)
yticks(-60:10:60)

```

```

xlabel('a*')
ylabel('b*')
legend({'3.1 real', '3.1 imaged', '3.1 matching', '3.2 real', '3.2 imaged', ...
       '3.2 matching'}, 'Location', 'southeast');

```

